

Information-Parity Testing for Code Representation Effects on Frontier LLMs: Methodology and Diagnostic Findings from a Bug-Detection Study

Takumi Shiraishi
takumi@wovol.com

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Abstract

Frontier LLMs are increasingly applied to code reasoning, yet whether the *format* of input code affects reasoning quality, independent of the *information content* of the representation, is rarely tested with controls. We present a methodology for such testing—a 5-condition design with an information-parity control (C1+) projecting a contract-augmented IR back into Python source—and instantiate it on T2 bug detection across 30 retained non-recursive Python functions, two confirmatory frontier LLMs, and a successor pair (post-confirmatory exploratory follow-on under a relaxed sampling regime, not a cross-generation replication). The primary endpoint H1 ($C4 > C1+$) yields a moderate positive effect (rank-biserial $r_{rb} = +0.581$, 95% CI $[+0.077, +0.917]$ excludes zero) but does not reach Holm-corrected significance under the pre-registered 9-test family (corrected $p = 0.205$). Exploratory analyses surface a directional pattern: H3 (in-text information enrichment helps) inverts on all four model-runs spanning two LLM generations. Post-hoc inspection reveals that the integer T2 composite score (range 0–3) saturates at 2—its ceiling of 3 is never attained—compressing the effective Wilcoxon n from 30 to 13–14 and constituting the structural source of confirmatory underpowering. We discuss implications for both the empirical claim and the methodology, and propose rubric refinement and T1/T3 collection as the next steps.

Contents

1	Introduction	4
1.1	The question	4
1.2	Why this is hard to test cleanly	4
1.3	Contributions	4
1.4	The single-sentence proposition	4
1.5	Headline empirical findings (preview)	5
2	Related Work	5
2.1	Code representation for LLMs (training-time and inference-time)	5
2.2	The closest ancestor: Namavar et al. (2022)	5
2.3	LLM prompt-format sensitivity	6
2.4	Neighbouring operational mechanisms	6
2.5	Novelty, circumscribed	6
3	Methodology	6
3.1	Study phases	6
3.2	Conditions and dataset	7
3.2.1	The five conditions	7
3.2.2	Dataset composition	7
3.3	Information parity: operational definition and construction	8
3.4	Tasks and primary endpoint	8
3.5	Models and sampling regime	9
3.6	Statistical plan	9
3.7	Score rubric and its empirical distribution	10
3.8	Pre-registration	10
4	Results	10
4.1	Confirmatory: H1/T2 primary endpoint	10
4.2	Confirmatory: secondary T2 layer and family completeness	11
4.3	Confirmatory: H1 on T1 (program understanding)	11
4.4	Confirmatory: H1 on T3 (code transformation), and a realised ceiling	12
4.5	Exploratory: H1 per-model direction concordance	13
4.5.1	Confirmatory pair, T2	13
4.5.2	Successor pair, T2 (under temperature relaxation)	13
4.6	Exploratory: H3 directional inversion across all four model-runs	14
4.7	Exploratory: cumulative-decomposition E1–E3	16
5	Discussion	16
5.1	What the primary endpoint does and does not support	16
5.2	Cross-run direction concordance: what it permits and forbids	17
5.3	The score-rubric ceiling is a structural finding, not a side issue	17
5.4	Implications for the long-term Lumen vision	17
5.5	Threats to validity (consolidated)	18

6 Conclusion	18
6.1 Summary	18
6.2 Immediate follow-on	19
6.3 Long-term	19
Appendices	21
A Retained 30 func_ids and domain categorization	21
B Pre-registration timeline and key commit hashes	21
C Full E1–E3 results, confirmatory v2 and successor runs	22
D Information-parity round-trip validation	23
E Reproducibility manifest	24

1 Introduction

1.1 The question

Frontier large language models are increasingly ubiquitous in developer workflows for code reasoning. Almost without exception, the input pipelines that feed them serialize code as plain source text, sometimes with light annotation. Whether plain text is the appropriate representation—and whether the *format* of the representation affects model reasoning *independently* of the information content it carries—is rarely tested with controls.

1.2 Why this is hard to test cleanly

The naive comparison—text against AST, say—confounds two variables. Two representations almost never carry exactly the same information; differences in annotations, derived edges, or type detail accompany differences in format, and any observed effect cannot be partitioned between the two axes. Prior work has either trained models on different representations (§2.1), confounding format with training signal, or applied structured prompting heuristically without controls (§2.3). To our knowledge no prior controlled study of code-representation effects on frontier LLMs combines a frozen-evaluation target, an information-parity control, and pre-registered family-wise correction. The remainder of this paper addresses that gap.

1.3 Contributions

This paper makes three contributions.

1. **A methodology for information-parity testing of code-representation effects on frozen frontier LLMs.** We define a five-condition design (§3.2.1) in which the central comparison is C4 against C1+: C1+ projects C4’s typed-and-contract-annotated intermediate representation back into Python source, holding information content constant against C4 and varying only format. The parity control (§3.3) isolates the structure axis from the information axis.
2. **A pre-registered confirmatory family with formal multiplicity control.** Three hypotheses crossed with three primary metrics constitute a nine-test family, corrected by Holm–Bonferroni at $\alpha = 0.05$ (§3.6). The statistical machinery was frozen before data collection.
3. **Diagnostic findings from a 30-function instantiation.** On a T2 bug-detection task we report a primary endpoint that is direction-consistent but Holm-non-significant (§4.1); an exploratory directional inversion of H3 replicated across all four model-runs (§4.6); and a structural source of underpowering—a near-saturated location sub-score that compresses the effective Wilcoxon sample size—that motivates rubric refinement before further empirical claims (§5.3).

1.4 The single-sentence proposition

We propose a controlled experimental methodology for measuring code-representation effects on frozen frontier LLMs, instantiate it on bug detection across 30 retained Python functions and two LLM generations, and report a primary endpoint that is direction-consistent but underpowered, alongside an exploratory directional inversion of the in-text-information-enrichment hypothesis and a structural source of underpowering inherent to the as-implemented scoring rubric.

1.5 Headline empirical findings (preview)

The primary endpoint on H1/T2 yields a rank-biserial $r = +0.581$ with 95% bootstrap CI $[+0.077, +0.917]$ and one-sided raw $p = 0.023$; under Holm correction at $m = 9$ the corrected $p = 0.205$, and the family-level decision is fail-to-reject (§4.1). The H1 direction is positive in all four model-runs across the confirmatory and successor pairs (§4.5). The H3 direction is negative in all four model-runs—opposite the pre-registered prediction (§4.6). No T2 record attains the composite ceiling of 3 (0/300 records; §3.7); the effective Wilcoxon n collapses to 13–14 on the central confirmatory cells. For this journal version the nine-cell family is completed: H1 (C4 over its information-parity control) is a *clean null* on T1 ($r = +0.10$, raw $p = 0.36$, no rubric ceiling) and ceiling-censored on T3 ($r = +0.43$, effective $n \approx 5$; 93.7% of scores at 1.0), so **no cell is rejected on any task** (§4.3–§4.4). The cleanest unconfounded datum, the T1 null, is evidence *against* a structure-at-constant-information advantage for program understanding.

Document scope. This paper is the extended preprint/journal-version draft, anticipated as an arXiv release under Constitution §14 (Phase 3 publishable outputs); a workshop-format abridgement, if pursued, would be prepared as a separate document and is not part of the present submission.

2 Related Work

2.1 Code representation for LLMs (training-time and inference-time)

Two lines of prior work bear on Lumen’s question.

Training-time integration of structure. Encoder and decoder models that bake code structure into their weights through specialized training objectives—CodeBERT (Feng et al., 2020), GraphCodeBERT (Guo et al., 2021), TreeBERT (Jiang et al., 2021), and InCoder (Fried et al., 2023)—learn from ASTs, data-flow graphs, or tree-positional encodings. Lumen does not retrain; it varies the input representation supplied to a frozen frontier LLM and treats representation format as a controlled independent variable.

Inference-time structured prompting. Chain-of-thought (Wei et al., 2022), structured-output schemas, and code-sketch scaffolds introduce structure heuristically at the prompt boundary. They adopt structure as a design choice; representation format is not varied as an independent variable under explicit controls.

2.2 The closest ancestor: Namavar et al. (2022)

The direct intellectual ancestor of this work is Namavar et al. (2022), who run a controlled comparison of code representations—token sequences, ASTs, and graphs—for learning-based program repair. Lumen sits in that lineage and differs along three axes.

Target. They train generative repair models from scratch on each representation; we evaluate frozen frontier LLMs and vary only the input representation. The shift reflects the 2022→2026 deployment context for LLM-based code reasoning rather than a deficiency in their design.

Information control. Their representations differ in both format and information content; Lumen adds C1+ (§3.3), an information-parity textual control that holds information content constant against C4 and varies only format. To our knowledge no prior controlled study of code-representation effects on frontier-LLM reasoning has used such a parity control.

Discipline. They do not pre-register a confirmatory family; Lumen pre-registers H1–H3 across three primary metrics and corrects the resulting nine-test family under Holm–Bonferroni at $\alpha = 0.05$ (§3.6).

2.3 LLM prompt-format sensitivity

A separate line of work documents that frontier LLMs are sensitive to surface-level prompt-format choices independent of the underlying information content (Sclar et al., 2024; Salinas and Morstatter, 2024). Lumen lifts that concern from prompt-design folklore into a domain-specific controlled study: the format axis is operationalized as a hierarchy of code representations (§3.2.1) and the information axis is held by the C1+ parity control (§3.3).

2.4 Neighbouring operational mechanisms

Three neighbouring areas are close enough to require separation from Lumen’s claim. Contract- and verifier-backed systems such as Clover (Sun et al., 2024) check consistency among generated code, specifications, and annotations. Deterministic AST analyses can likewise detect and correct hallucinations in generated code (Khatri et al., 2026). Both are operational-consistency mechanisms: they filter or repair generated artifacts, but they do not vary representation format under information control.

Focused reasoning prompts occupy a different boundary. Chain-of-thought prompting (Wei et al., 2022), Plan-and-Solve prompting (Wang et al., 2023), and program-aided language models (Gao et al., 2023) alter the form of the reasoning interface rather than the information content of the code representation. PAL further delegates execution to an external interpreter, placing its gains on the capacity/operational side rather than the representation side. These lines therefore reinforce the circumscribed novelty claim below: Lumen’s contribution is not structure, checking, or prompting alone, but their controlled comparison under an information-parity control.

2.5 Novelty, circumscribed

The contribution claimed here is methodological: the information-parity control and the pre-registered, family-wise corrected confirmatory design are themselves the deliverable (§1.3), not the outcome of any single instantiation. The novelty is the controlled *combination*—information-parity testing of frontier-LLM code reasoning under pre-registered family-wise correction—rather than any individual ingredient. ASTs, type annotations, contract-style specifications, frozen-LLM evaluation, and pre-registration each pre-exist; the novelty rides on the combination and on the control. The 30-function T2 instantiation reported below is diagnostic of that instrument, not a referendum on it: the fail-to-reject outcome (§4.1) bounds what this dataset demonstrates while leaving the methodological contribution intact, and it is reported in full precisely because the method, not a positive result, is what the paper offers. Long-term inspirations from content-addressed code storage and projectional editing inform the broader Lumen research program but are out of scope for the present submission.

3 Methodology

3.1 Study phases

The study was conducted over a 90-day window (Constitution §13) in three phases.

Phase 1 (days 1–25) covered pipeline construction and dataset curation. We implemented the representation transformers, the contract generator, and the automated scoring scripts, and curated the 30 retained non-recursive Python functions. This phase produced the conversion pipelines for C1 through C4 and the C4 → C1+ projection.

Phase 2 (days 26–50) was a pilot. A pilot run exercised a small subset—approximately three functions across all five conditions, all three tasks, and both models, on the order of ninety items per run—to calibrate the runner and the automated scoring scripts. Two pilot runs were executed; the second was clean. Pilot data is not pooled with confirmatory data: the confirmatory analysis derives solely from a fresh collection on the 30 retained functions (`full_t2_confirmatory_v2`).

Constitution v0.2.2 §16 (Open Question 4) anticipated a Phase-2 pilot power analysis or effect-size sanity check to decide whether 30 functions suffice or expansion to 50 is needed; no such artifact was produced, and the dataset was advanced to confirmatory collection at the constitutional minimum line of $n = 30$. The score-rubric ceiling documented in §3.7 and analyzed in §5.3 was identified post hoc from the confirmatory data; the pre-confirmatory dataset-sufficiency audit that records this gap is registered in Appendix E.

Phase 3 (days 51–90) covered confirmatory data collection (T2 only), a post-confirmatory exploratory replication on a successor frontier model pair, and analysis.

3.2 Conditions and dataset

3.2.1 The five conditions

Table 1 summarizes the five experimental conditions. The cumulative chain $C1 \rightarrow C2 \rightarrow C3 \rightarrow C4$ progressively enriches information, while the branch $C1 \rightarrow C1+$ enriches information in textual form. The central comparison is $C4$ versus $C1+$: both carry the same type and contract annotations and differ only in format—structured IR versus annotated Python source. This pairing isolates the structure axis from the information axis. The five conditions and their two enrichment axes are shown schematically in Figure 1.

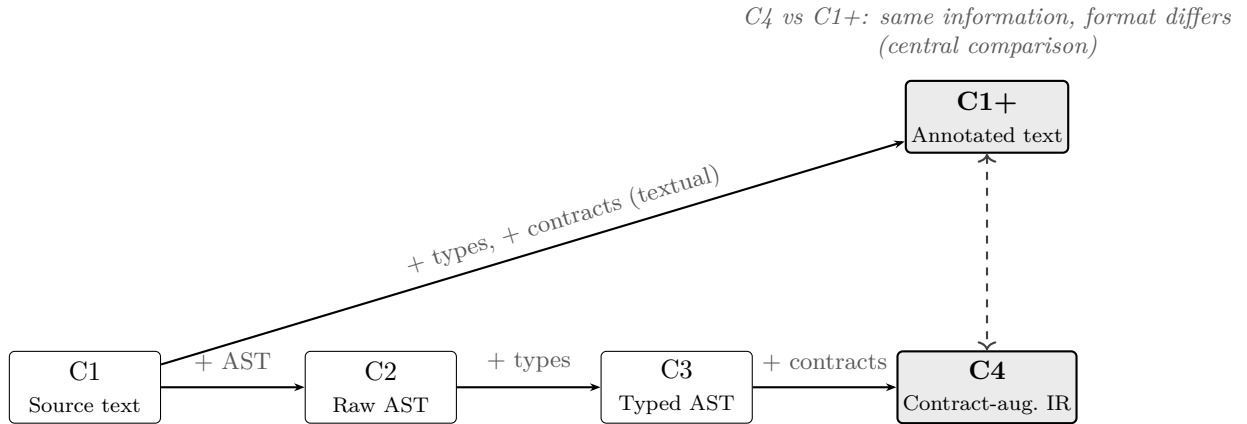


Figure 1. Schematic of the five conditions: cumulative chain $C1 \rightarrow C2 \rightarrow C3 \rightarrow C4$ and information-parity branch $C1 \rightarrow C1+$. (Optional supplementary figure.)

Table 1. The five experimental conditions. Information content is held constant between $C1+$ and $C4$ by construction; format differs (textual vs structured).

Cond	Label	Format	Info	Purpose
C1	Source text	Text	Base	Baseline
C1+	Annotated text	Text	Enriched	Information-parity control
C2	Raw AST	Structured	Base	Pure structure
C3	Typed AST	Structured	+ Types	Type effect
C4	Contract-augmented IR	Structured	+ Contracts	Full enrichment, structured

3.2.2 Dataset composition

The dataset consists of 30 retained non-recursive single Python functions, manually curated within the 90-day scope. Their domain distribution (Appendix A) is predominantly list and array manipulation (20 of 30), with smaller representation from descriptive statistics (6), string processing

(2), arithmetic (1), and control-flow or parser-like logic (1). This list-operations predominance reflects the prevalence of such functions in standard-library and algorithmic-task settings rather than intentional curation; we treat it as an external-validity limitation (§5.5).

A static recursion-exclusion check at the curation stage rejects any function with self-recursion or mutually recursive helpers; the 90-day scope restricts the study to non-recursive code (Constitution §17.5).

For each function we prepare ground-truth artifacts: a single buggy variant whose defect falls into one of five categories (off-by-one, wrong comparison, missing boundary, incorrect variable, swapped arguments; Constitution §17.3), an annotation of the bug location and category, and a reference-fix test suite.

3.3 Information parity: operational definition and construction

We define the *information content* of a representation as the set of typed propositions derivable from it: the function signature types and the parameter and return contracts (precondition, post-condition, invariant). Under this definition C1+ and C4 are equivalent by construction.

C1+ is produced by projecting C4’s annotations into Python source: PEP 484 inline type hints in the signature, plus a structured docstring block carrying the contract clauses in a fixed template. The projection is implemented in `src/pipeline/ir_to_annotated_text.py`.

We acknowledge a residual asymmetry. The projection does not preserve AST node-kind metadata, IR-level syntactic child-ordering, or explicit cross-reference structure among contract clauses, which become free-form docstring text. Under our operational definition these are *encoding*, not *information*; a more demanding definition—Shannon-style bit count, or recoverability of the source IR from the projected text—would assign C4 strictly higher information content. We do not adopt that definition, because the empirical question is whether structure aids reasoning *given the same propositional content*, not whether representations differ in arbitrary encoding bits. A round-trip validation measuring $C4 \rightarrow C1+ \rightarrow C4'$ structural recovery (Appendix D) quantifies this residual asymmetry: across the 30 functions the propositional content is recovered essentially in full (contract clauses at 100%, the shared-identifier links between clauses at 100%, AST node kinds at 99.8%, type annotations at 96.4%, parent-child structure at 98.7%), while byte-exact AST identity is never recovered (0/30). The non-preservation is therefore concentrated in syntactic encoding—child-ordering and node-level metadata, and the explicit cross-reference *structure* that becomes free-form docstring text—whereas the referential information it carried (which clauses mention which variables) remains recoverable. This refines the qualitative disclosure above rather than overturning it.

Contracts for C4, and therefore for C1+, are generated by an LLM that is not used as a test subject and are reviewed by the author under the contract review protocol (Constitution §17.2): the allowed edits are factual correction, deletion of vacuously true clauses, and a single missing-clause addition. Provenance—raw, reviewed, and diff—is logged for every function.

3.4 Tasks and primary endpoint

Three task types are defined: T1 (code understanding), T2 (bug detection), and T3 (code transformation). T1 asks the model to summarize a function’s behavior in natural language from its representation; T3 asks the model to produce a modified version of a function satisfying a contract-described change. Both are constitutional task families; the 90-day study collected only T2. T1 and T3 were collected afterward for this journal version under separate pre-registrations (t1-v1, t3-v1), completing the nine-cell family (§4.3, §4.4).

T2 was pre-registered as the primary endpoint (Constitution §17.4) for objectivity: its 0–3 composite score—the sum of location, diagnosis, and fix sub-scores—is fully automated and requires

no human judgment at scoring time. We honor that pre-registration throughout §4: H1 on T2 is the primary endpoint and carries the strongest interpretive weight.

3.5 Models and sampling regime

The confirmatory pair comprised `gpt-5.4` and `claude-opus-4-6`, sampled at `temperature=0`. A post-confirmatory exploratory replication used the same-provider successor frontier pair—`gpt-5.5` and `claude-opus-4-7`—sampled at provider-default temperature. The successor models reject `temperature=0` at the API level; provider-default temperature is therefore a hard constraint, not a methodological preference.

This introduces a sampling-regime asymmetry: deterministic on the confirmatory pair, stochastic on the successor pair. We accordingly frame the cross-pair comparison as a robustness check under temperature relaxation, not as cross-generation replication; the two pairs are not directly comparable under a shared sampling regime. The successor pair was selected as the same-provider frontier successors so that representation effect, rather than provider drift, remains the axis of interest.

A brief disambiguation on terminology. The successor run is called a “replication” in the abstract and in §3.1 as a description of the *process*: a fresh run on a successor model pair, identified in the run logs as `t2_frontier_successor_replication`. What this subsection denies is “replication” as an *inferential framing* of the cross-pair evidence: the deterministic/stochastic asymmetry above precludes that framing, and §4.5 returns to the point. Both usages are correct; they refer to different things.

3.6 Statistical plan

The pre-registered confirmatory family comprises three hypotheses (H1: $C4 > C1+$; H2: $C4 > C1$; H3: $C1+ > C1$) crossed with three primary metrics (T1, T2, T3), giving nine tests. For each test we compute, per function, a paired difference between the two conditions after averaging the two model scores; we then apply the Wilcoxon signed-rank test with a one-sided “greater” alternative matching the pre-registered direction. The effect size is the rank-biserial correlation

$$r_{\text{rb}} = \frac{T^+ - T^-}{T^+ + T^-},$$

where T^+ and T^- are the sums of the positive and negative signed ranks; we use the abbreviation r_{rb} throughout §4 and §5. We apply the Holm–Bonferroni step-down correction across the nine-test family at $\alpha = 0.05$. Holm–Bonferroni is a family-wise-error-rate correction that orders the raw p -values from smallest to largest and tests each in turn against a tightening threshold (α/m at the smallest rank, $\alpha/(m-1)$ at the next, and so on up to α at the largest); a rank is rejected only if it and every earlier rank clear their thresholds. We also report a 95% percentile bootstrap CI on r_{rb} from 10,000 seeded resamples.

We honor $m = 9$ as pre-registered. In the preprint, six of the nine cells were insufficient-data placeholders; for this journal version T1 and T3 were collected (§4.3–§4.4) and all nine cells are now tested, with $m = 9$ retained throughout. We did not at any stage retroactively reduce m to 3. Post-hoc m -reduction would forfeit the value of pre-registration: the cost is reduced corrected power, the benefit is that the result speaks under the rules committed to before data were seen. Effective sample size per test is further compressed because Wilcoxon discards zero paired differences: against $n_{\text{pairs}} = 30$, the tie distribution induced by the rubric (§3.7) leaves an effective n of 13–14. The statistical plan also fixes the exploratory/confirmatory boundary so that downstream sections can quote it without re-deciding: per-model analyses are exploratory by Constitution §17.4 and are reported uncorrected with explicit exploratory labeling.

On bootstrap CI provenance: Constitution §17.4 mandates 95% bootstrap CIs on effect sizes. The frozen `analyze_confirmatory.py` does not compute CIs; CIs were computed via a sibling implementation verified against the frozen analyzer’s point estimates. CI computation does not alter any hypothesis-testing decision; it adds reporting fidelity. Confirmatory CIs are reported in §4 alongside point estimates; provenance is recorded in Appendix E.

3.7 Score rubric and its empirical distribution

The T2 composite score is the integer sum of three binary sub-scores: location (0/1), diagnosis (0/1), and fix (0/1). Per-model support is therefore $\{0, 1, 2, 3\}$; the per-(function, condition) 2-model-averaged support is $\{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$.

We surface the empirical distribution here—before results—because it constrains what the analysis can detect. The confirmatory run contains 300 per-model T2 records (30 functions $\times$ 2 models $\times$ 5 conditions); their raw composite distribution is 0: 19 records (6%), 1: 89 (30%), 2: 192 (64%), and 3: 0 (0%). No model scored 3/3 on any T2 item; the location sub-score is the empirical bottleneck. After 2-model averaging, composite supports 2.5 and 3.0 are both empty, leaving an effective range of $[0.0, 2.0]$ with mode 2.0. Across H1, H2, and H3 at least 53% of paired differences are exactly zero; Wilcoxon discards these, and the effective n contracts to 13–14 from $n_{\text{pairs}} = 30$.

The ceiling is a property of the as-implemented rubric, not of the data. The location sub-score’s saturation rate is a per-item property: each function independently lands somewhere in the $\{0, 1, 2, 3\}$ raw composite distribution shown above, and that distribution is determined by what the rubric can score, not by how many functions are sampled. Expanding n alone therefore does not solve underpowering—additional functions arrive drawn from the same distribution and produce zero-valued paired differences at the same rate, so the effective Wilcoxon n stays in the same fraction of n_{pairs} . The path forward (§6) is rubric refinement, or supplementing the composite with a continuous-valued T2 metric. Score histograms are shown in Figure 4.

3.8 Pre-registration

The Lumen Design Constitution v0.2.2—including the full statistical plan—was committed to the public repository before data collection began (commit `6b83bca`). The constitutional document at `docs/design-constitution.tex` was committed at repository scaffolding and has not been edited since, which strengthens the pre-registration claim against post-hoc adjustment. The remaining pre-registration timeline—analysis-machinery freeze, confirmatory closure, successor-plan registration, and exploratory tooling—is recorded with commit hashes in Appendix B.

One naming artifact warrants explicit note alongside this provenance. The runner module is `src/experiment/run_pilot.py`, named during the Phase 2 pilot and reused without modification for confirmatory data collection. The name reflects implementation history, not methodological status: the confirmatory run is not a pilot.

4 Results

4.1 Confirmatory: H1/T2 primary endpoint

The primary endpoint of the study is the pre-registered H1 test on T2: under the C4-versus-C1+ comparison, does the structured-IR condition outperform its information-parity textual control on bug-detection composite score? Table 2 reports the full result.

Decision: fail to reject H_0 at $\alpha = 0.05$ corrected. With H2/T2 and H3/T2 also non-rejected at the family-corrected level (§4.2), the confirmatory family yields no rejection on T2. Under Constitution §9 the study therefore falls on the *Informative Negative Outcomes* path rather than meeting the Minimum-success threshold, which requires at least one of H1–H3 to be rejected

at the corrected significance level. The materials are publicly released in accordance with the same criterion.

The non-rejection is informative rather than empty. The point estimate $r_{rb} = +0.581$ is moderate and lies in the pre-registered direction; the 95% percentile bootstrap CI $[+0.077, +0.917]$ excludes zero; and the one-sided raw $p = 0.023$ would reject H_0 in isolation. The failure occurs at the Holm–Bonferroni family-correction stage with $m = 9$: the corrected $p = 0.205$, and at the present effective Wilcoxon $n = 14$ (compressed from $n_{\text{pairs}} = 30$ by the rubric-induced tie distribution) the correction absorbs the moderate effect. The structural source of the under-rejection—the score-rubric ceiling—is documented in §3.7 and discussed in §5.3.

Table 2. H1/T2 primary-endpoint (*Confirmatory analysis*). Bootstrap parameters: seed 20260427, 10 000 resamples, percentile method. The 95% CI on r **excludes zero**.

Field	Value
Wilcoxon W	83.0
Raw p (one-sided “greater”)	0.023
Holm-corrected p ($m = 9$)	0.205
Rank-biserial r	+0.581
95% bootstrap CI on r	$[+0.077, +0.917]$
n_{pairs}	30
$n_+ / n_- / n_=$	10 / 4 / 16
Effective Wilcoxon n	14
Median paired difference	0.0
Mean paired difference	+0.183
Decision: fail to reject H_0 at $\alpha = 0.05$ corrected.	

4.2 Confirmatory: secondary T2 layer and family completeness

Table 3 reports the full nine-test confirmatory family: three hypotheses crossed with three primary metrics, with $m = 9$ honored as pre-registered.

Of the three tested cells (all on T2), two yield rank-biserial r consistent with their pre-registered direction: H1/T2 with $r_{rb} = +0.581$ (CI $[+0.077, +0.917]$, raw $p = 0.023$) and H2/T2 with $r_{rb} = +0.200$ (CI $[-0.333, +0.700]$, raw $p = 0.233$). The third, H3/T2, yields $r_{rb} = -0.473$ (CI $[-0.885, +0.077]$, raw $p = 0.969$)—*opposite* the pre-registered direction. None of the three reaches Holm-corrected significance. The H3/T2 upper bound ($+0.077$) only just crosses zero.

The family is now complete. The T1 family was collected under pre-registration **t1-v1** (§4.3) and the T3 family under **t3-v1** (§4.4); all nine cells are tested. **No cell is rejected at the Holm-corrected level:** the smallest raw p in the family is H1/T2’s 0.023, which already fails the $m = 9$ Holm threshold ($0.05/9 = 0.0056$), so every cell corrects to Holm $p \in \{0.205, 1.000\}$. The pre-registered Informative-Negative-Outcomes classification (Constitution §9) therefore stands across all three tasks, not T2 alone.

The H3/T2 directional inversion is the strongest exploratory signal in the dataset; its per-model decomposition is taken up in §4.6.

4.3 Confirmatory: H1 on T1 (program understanding)

The T1 family was collected after the preprint under pre-registration **t1-v1**, using the frozen deterministic checklist scorer over all 30 confirmatory functions and the frozen model pair (complete: 300/300 cells, both models in every condition). The central test **H1/T1 (C4 vs C1⁺)** yields $r_{rb} = +0.098$ (raw one-sided $p = 0.361$; 95% bootstrap CI $[-0.440, +0.655]$), an effect indistinguishable from zero with a CI that spans the origin. The secondary T1 cells point weakly in the pre-registered

Table 3. Confirmatory 9-test family, now complete: all nine cells tested (T2; T1 via `t1-v1`; T3 via `t3-v1`) (*Confirmatory analysis*). $m = 9$ Holm correction as pre-registered. **No cell is rejected at the Holm-corrected level.** The three T3 cells are ceiling-censored (93.7% of T3 scores at 1.0; see §4.4).

Hyp	Task	n	W	raw p	Holm p ($m = 9$)	r_{rb}	95% CI	Status
H1	T1	30	84.0	0.361	1.000	+0.098	$[-0.440, +0.655]$	tested
H1	T2	30	83.0	0.023	0.205	+0.581	$[+0.077, +0.917]$	tested
H1	T3	26	15.0	0.219	1.000	+0.429	$[-1.000, +1.000]$	tested (ceiling)
H2	T1	30	80.5	0.259	1.000	+0.184	$[-0.429, +0.684]$	tested
H2	T2	30	63.0	0.233	1.000	+0.200	$[-0.333, +0.700]$	tested
H2	T3	26	0.0	1.000	1.000	-1.000	$[-1.000, +0.000]$	tested (ceiling)
H3	T1	30	92.0	0.388	1.000	+0.076	$[-0.453, +0.595]$	tested
H3	T2	30	24.0	0.969	1.000	-0.473	$[-0.885, +0.077]$	tested
H3	T3	26	1.0	0.938	1.000	-0.800	$[-1.000, +1.000]$	tested (ceiling)

direction—H2/T1 (C4 vs C1) $r_{\text{rb}} = +0.184$ (raw $p = 0.259$) and H3/T1 (C1⁺ vs C1) $r_{\text{rb}} = +0.076$ (raw $p = 0.388$)—but neither approaches significance, and all three correct to Holm $p = 1.000$ in the nine-test family. **On the understanding task the structure-at-constant-information advantage is absent:** C4 does not outperform its information-parity textual control. The realised score distribution is graded (300 scored cells, 9 distinct values, mean 0.900, 58.7% at the 1.0 ceiling)—right-skewed but free of the three-level collapse that compressed T2.

Data-collection note (pre-registered missing-data handling). On the first collection pass, 27 of the 300 T1 calls failed, all on `claude-opus-4-6`, because the Anthropic API credit balance was exhausted mid-run (`gpt-5.4`: zero failures). The run is resumable; after a credit top-up it was resumed and all 27 cells were recovered, so the *final* T1 dataset is complete (300/300, zero failures) and no missing-data imputation is required—the pre-registered complete-case and worst-case arms (`t1-v1` §5.2) coincide ($n = 30$, $r_{\text{rb}} = +0.098$). We record the transient gap for provenance; it does not affect the reported result. Per-cell status counts are archived at `results/analysis/exploratory/full_t1_confirmatory_v1/sensitivity.json`.

4.4 Confirmatory: H1 on T3 (code transformation), and a realised ceiling

The T3 family was collected under pre-registration `t3-v1`, using the frozen test-pass-fraction scorer executed in a sandboxed subprocess with a 10s wall-clock timeout. Of the 30 confirmatory functions, 26 carry a frozen post-transform suite; the other four (`peak_valley_indices`, `segments_above_threshold`, `stable_partition`, `group_by_key`) have no T3 ground truth and were pre-declared as structural exclusions before any T3 call.

H1/T3 (C4 vs C1⁺) yields $r_{\text{rb}} = +0.429$ in the pre-registered direction but with raw $p = 0.219$ and a 95% CI of $[-1.000, +1.000]$ —uninformative. The reason is the *realised ceiling* pre-registered as the primary T3 threat (`t3-v1` §3.3): 93.7% of T3 scores are exactly 1.0 (mean 0.961, 7 distinct values over 255 scored cells). Both frozen models solve almost every transformation in almost every representation, so the paired C4–C1⁺ differences are overwhelmingly ties: only ~ 5 of the 26 functions have a non-zero difference (effective Wilcoxon $n \approx 5$). The test cannot reject, not because the effect is bounded away from the pre-registered direction, but because the top-of-scale saturation destroys resolution—the mirror image of the T2 ceiling (paper §3.7), and exactly the failure mode anticipated at pre-registration. The secondary cells inherit the same collapse: H2/T3 $r_{\text{rb}} = -1.000$ (effective $n = 1$) and H3/T3 $r_{\text{rb}} = -0.800$ (effective $n = 3$) are computed on a handful of non-tied pairs and carry no interpretive weight. All three correct to Holm $p = 1.000$.

Failure handling and sensitivity (`t3-v1` §5). Only 5 of 260 cells were parse/execution failures (`claude-opus-4-6`: C1 1, C2 1, C4 2; `gpt-5.4`: C3 1)—no MNAR concentration. The

frozen `analyze_confirmatory.py` treats non-ok cells as missing (per-function mean over present models), so the pre-registered “retain-as-0.0” rule is realised as the worst-case arm; with 5/260 cells affected this is immaterial. Worst-case imputation gives H1/T3 $r_{rb} = +0.286$ (raw $p = 0.289$), preserving the (non-significant) positive direction. Artifacts: `results/analysis/exploratory/full_t3_confirmatory_v1/sensitivity.json`.

Reading across the completed family. The central Lumen test, H1 (C4 over its information-parity control), is *not* supported on any task: it is a clean null on T1 ($r_{rb} = +0.10$), a moderate but multiplicity-absorbed positive on T2 ($r_{rb} = +0.58$), and ceiling-censored on T3 ($r_{rb} = +0.43$, effective $n \approx 5$). Two of the three tasks are measurement-limited (T2 by a low-end rubric ceiling, T3 by a high-end solve-rate ceiling), so the completed family constrains the structure-at-constant-information claim less than its nine cells suggest; the cleanest single datum is the T1 null. This is taken up in the decision-gate evaluation (`docs/next-phase-design.md §4`).

4.5 Exploratory: H1 per-model direction concordance

The analyses in this and the following two subsections are exploratory by Constitution §17.4; they are reported uncorrected and are not elevated to confirmatory claims. The two per-model panels below report H1/T2 on the confirmatory and successor pairs in turn; their joint reading is then summarized.

4.5.1 Confirmatory pair, T2

Table 4 reports H1/T2 split by model on the confirmatory pair. `claude-opus-4-6` yields $r_{rb} = +0.750$ with 95% CI $[+0.142, +1.000]$ excluding zero (raw one-sided $p = 0.055$); `gpt-5.4` yields $r_{rb} = +0.394$ with CI $[-0.250, +0.864]$ (raw $p = 0.135$). Both per-model estimates point in the pre-registered direction.

Table 4. H1/T2 per-model rank-biserial, confirmatory pair (*Exploratory analysis*, uncorrected).

Model	n	r_{rb}	95% CI	raw p (1-sided)
<code>claude-opus-4-6</code>	30	+0.750	[+0.142, +1.000]	0.055
<code>gpt-5.4</code>	30	+0.394	$[-0.250, +0.864]$	0.135

4.5.2 Successor pair, T2 (under temperature relaxation)

Table 5 reports the same split on the successor pair, sampled at provider-default temperature (§3.5). `claude-opus-4-7` yields $r_{rb} = +0.509$ (CI $[-0.143, +1.000]$, raw $p = 0.093$); `gpt-5.5` yields $r_{rb} = +0.600$ (CI $[-0.333, +1.000]$, raw $p = 0.188$). The `gpt-5.5` row is computed on $n = 26$: twelve parse-failure exclusions in the successor run reduced its effective pair count, distributed across non-C1 conditions.

Table 5. H1/T2 per-model rank-biserial, successor pair (*Exploratory analysis*, uncorrected). Provider-default temperature. `gpt-5.5` $n = 26$ due to 12 parse-failure exclusions in the successor run, distributed across non-C1 conditions.

Model	n	r_{rb}	95% CI	raw p (1-sided)
<code>claude-opus-4-7</code>	30	+0.509	$[-0.143, +1.000]$	0.093
<code>gpt-5.5</code>	26	+0.600	$[-0.333, +1.000]$	0.188

Across all four model-runs the H1 direction is consistent (per-model effect sizes and CIs in Tables 4 and 5). The successor pair preserves the direction under a relaxed sampling regime, so the H1 direction is not a single-model artifact and is not erased by stochastic sampling. We frame this as

robustness evidence under temperature relaxation; the deterministic-versus-stochastic asymmetry between the two pairs (§3.5) precludes a cross-generation-replication framing. Figure 2 renders the four rows as a forest plot with a vertical zero line.

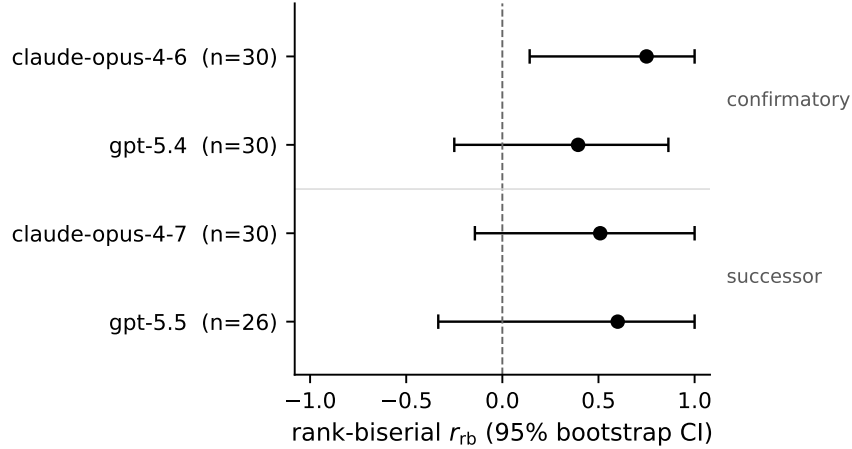


Figure 2. Forest plot of H1/T2 per-model rank-biserial with 95% CI across the confirmatory and successor pairs (4 rows). Vertical zero line indicated. *Exploratory analysis.*

4.6 Exploratory: H3 directional inversion across all four model-runs

All four model-runs show a negative H3 direction (with the caveat that the `claude-opus-4-7` row’s r_{rb} is at the estimator’s lower bound under a collapsed CI; see the in-line caveat below and the table footnote, and read the four-run count as direction sign only). The pre-registered prediction was $C1+ > C1$ —that information enrichment in textual form would aid T2—but Table 6 shows $C1+$ underperforming $C1$ in every run: `claude-opus-4-6` $r_{rb} = -0.333$, `gpt-5.4` -0.400 , `claude-opus-4-7` -1.000 , and `gpt-5.5` -0.500 .

The `claude-opus-4-7` row warrants a caveat in line. Its $r_{rb} = -1.000$ with collapsed 95% CI $[-1.000, -1.000]$ reflects 0 positive against 8 negative paired differences in 30 pairs (22 ties); under such sparse non-zero data the rank-biserial estimator saturates at its lower bound and the CI degenerates. This is an artifact of the estimator at small effective n , not a strong negative claim.

Treating each model-run’s H3 sign as a single observation under a null of no effect gives $(1/2)^4 = 1/16 \approx 0.063$ for four negatives out of four. We report this as an upper-bound exploratory consistency check, not as an inferential test: the four runs share dataset and rubric, and the two members of each pair share inputs, so the runs are not strictly independent and the bound is loose.

An auxiliary observation from parse logs: `gpt-5.5` produced unparseable responses on 0/30 $C1$ items in the successor run but on 12/120 non- $C1$ items; among the non- $C1$ conditions it is the structured and enriched ones that destabilize the model’s output. The instability is concentrated where the input departs from plain source.

This concentration also explains the per-pair n asymmetry between Table 5 and Table 6 on the `gpt-5.5` row. H1 compares $C4$ to $C1+$, both of which are non- $C1$ and so vulnerable to parse failure, and the row drops to $n = 26$; H3 compares $C1+$ to $C1$, and $C1$ always parses, so only $C1+$ failures reduce the pair count and the row stands at $n = 28$.

Parse-failure baseline and worst-case sensitivity. The confirmatory pair (`gpt-5.4`, `claude-opus-4-6`) had *zero* parse failures across all 300 T2 items (0/300); the 12 failures are exclusive to the successor

model **gpt-5.5** (12/120 of its items, 0 for **claude-opus-4-7**), and entirely in non-C1 conditions (C1+ 2, C2 4, C3 3, C4 3). Because the exclusion is missing-not-at-random—failure correlates with the representation condition—we ran a worst-case sensitivity analysis imputing every parse-failed item at the lowest score (0) rather than excluding it. Under worst-case imputation the H3 inversion is unchanged in direction for all four runs and, for the affected **gpt-5.5** row, *strengthens* from $r_{rb} = -0.500$ to -0.762 (imputing the two C1+ failures at 0 further depresses C1+ relative to C1); the **claude-opus-4-7** row is unaffected (no C1+ failures). The directional finding therefore does not depend on the exclusion rule. Per-cell counts and the full exclude-versus-impute table are reported as exploratory analysis in `results/analysis/exploratory/r2_2_parse_failure_sensitivity/`.

Taken together, the data exploratorily suggest a pattern about in-text annotation. On top of plain Python source, in-text type and contract annotations do not aid—and may impede—frontier LLM bug detection. The qualifier “at the current scoring rubric” is essential to the reading: the pattern is observed under the as-implemented T2 composite (§3.7), and a refined rubric could reposition the comparison. The pattern is consistent with, but does not confirm, the dissociation of representational structure from information content that the C1+ control was designed to test. Figure 3 renders the four rows as a forest plot and complements Figure 2.

Table 6. H3/T2 per-model rank-biserial across all four model-runs (*Exploratory analysis*). All four directions are negative. The **claude-opus-4-7** row has a collapsed CI at $[-1.000, -1.000]$ reflecting 0 positive vs 8 negative diffs in 30 pairs (22 ties); this is a CI artifact at small effective n , not a strong negative claim. Parse-failure baseline: the confirmatory pair had 0/300 failures; the **gpt-5.5** successor row reflects 12/120 non-C1 failures, and the direction survives worst-case (0-imputation) sensitivity (§4.6).

Run	Model	n_{pairs}	r_{rb}	95% CI	$n_{+}/n_{-}/n_{=}$
confirmatory v2	claude-opus-4-6	30	-0.333	$[-1.000, +0.333]$	3/6/21
confirmatory v2	gpt-5.4	30	-0.400	$[-1.000, +0.333]$	3/6/21
successor	claude-opus-4-7	30	-1.000	$[-1.000, -1.000]$	0/8/22
successor	gpt-5.5	28	-0.500	$[-1.000, +1.000]$	1/3/24

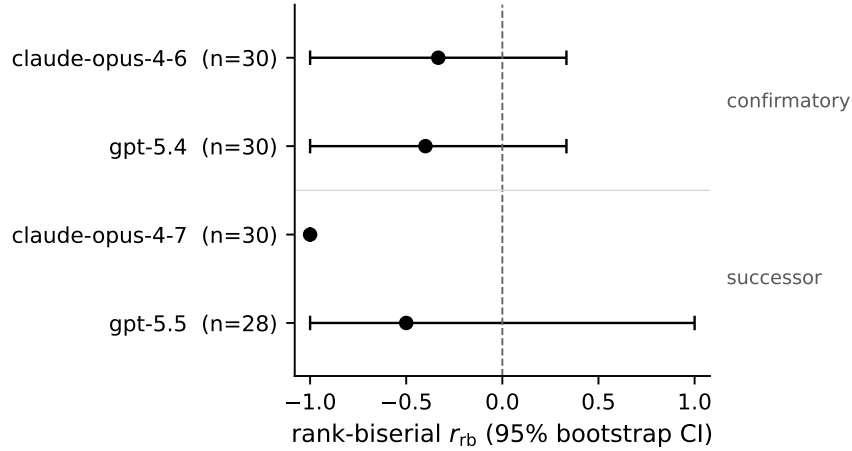


Figure 3. Forest plot of H3/T2 per-model rank-biserial with 95% CI across the confirmatory and successor pairs (4 rows). Vertical zero line indicated. Demonstrates the inverted-direction pattern visually; complements Figure 2. *Exploratory analysis*.

4.7 Exploratory: cumulative-decomposition E1–E3

Table 7 reports the cumulative-decomposition effect sizes for E1 (C2 vs C1), E2 (C3 vs C2), and E3 (C4 vs C3) on both runs; the full numerical table—with W , raw p , CI, and tie counts—is in Appendix C. On the confirmatory pair, r_{rb} values are E1 +0.250, E2 −0.200, E3 +0.250; on the successor pair, E1 −0.341, E2 +0.250, E3 −0.333.

No single decomposition step reaches significance in either run, and the direction inverts between the confirmatory and successor runs on E1 and E3. The cumulative chain $C1 \rightarrow C2 \rightarrow C3 \rightarrow C4$ does not exhibit a clean monotone build-up of effect through the four conditions.

This pattern, observed post hoc, is consistent with the pre-registered choice of central comparison: the systematic signal in the data is carried by the structure-versus-information axis (C4 vs C1+, isolated by construction) rather than by the marginal information additions along the cumulative chain.

Table 7. Cumulative-decomposition per-pair effect sizes (*Exploratory analysis*, 2-model averaged). Full numerical table in Appendix C.

Run	Hyp	Pair	r_{rb}	Direction
v2	E1	C2 vs C1	+0.250	+
v2	E2	C3 vs C2	−0.200	−
v2	E3	C4 vs C3	+0.250	+
successor	E1	C2 vs C1	−0.341	−
successor	E2	C3 vs C2	+0.250	+
successor	E3	C4 vs C3	−0.333	−

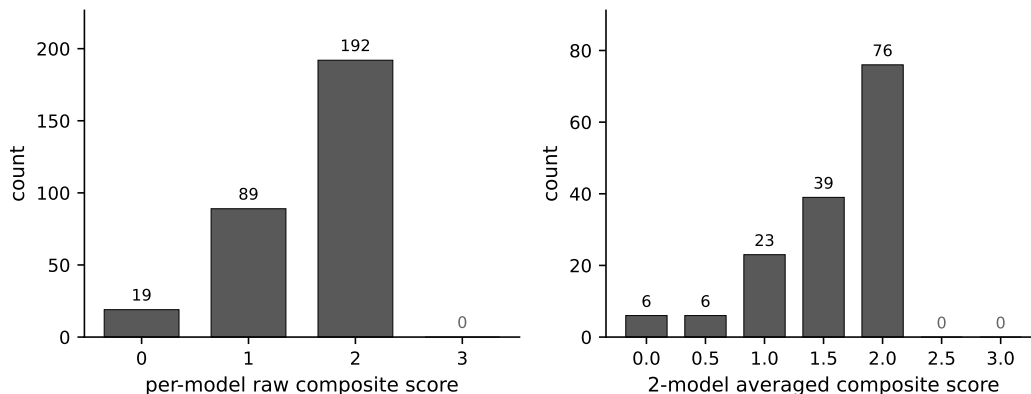


Figure 4. Score distribution histograms. Left: per-model raw composite $\{0, 1, 2, 3\}$ (composite 3 absent in 300/300 records). Right: per-(function, condition) 2-model-averaged composite $\{0, 0.5, \dots, 3.0\}$ over all $30 \times 5 = 150$ cells (supports 2.5 and 3.0 both absent across all 150 cells). Visual support for §3.7.

5 Discussion

5.1 What the primary endpoint does and does not support

The non-rejection of H1/T2 at the Holm-corrected family level (§4.1) is not evidence of no effect. The honest reading of the primary endpoint splits across two clauses that must hold together; neither overrides the other.

First clause: the point estimate. The point estimate $r_{rb} = +0.581$ is moderate and lies in the pre-registered direction, and the one-sided raw $p = 0.023$ would reject H_0 in isolation. The data are consistent with the pre-registered direction at moderate effect size.

Second clause: the family correction. The failure occurs at the Holm–Bonferroni family-correction stage with $m = 9$ (§3.6), six of whose cells are insufficient-data placeholders contributing Holm $p = 1.000$ by convention; the corrected $p = 0.205$; and the effective Wilcoxon $n \approx 14$, compressed from $n_{\text{pairs}} = 30$ by the rubric-induced tie distribution, does not support a confirmatory claim.

The CI excluding zero qualifies the non-rejection; it does not overturn it.

A back-of-the-envelope point reinforces the second clause. At the most-promising rank in $m = 9$, the Holm-corrected α is $\alpha/m = 0.05/9 \approx 0.0056$; detecting $r \approx +0.58$ at that threshold would require substantially larger non-tied effective n than the present data provide. The implication, taken up in §5.3 and §6.2, is that rubric refinement, not n alone, is the path forward.

5.2 Cross-run direction concordance: what it permits and forbids

The H1 direction is positive in all four model-runs across the confirmatory and successor pairs (§4.5); the H3 direction is negative in all four (§4.6). These analyses are exploratory under Constitution §17.4, uncorrected, and underpowered per cell, and cannot be promoted to confirmatory claims.

What the concordance *permits*. It constrains the explanation space. The positive H1 direction is not a single-model artifact, and it survives a relaxed sampling regime on the successor pair. The H3 inversion is not noise on one run; it appears in every run. Together these patterns shift the prior on what a properly powered confirmatory study would find, in opposite directions for the two hypotheses. What the concordance *forbids*. It does not promote H1 to confirmatory status, and it does not establish the H3 inversion as a genuine direction. Multiplicity is unaddressed; the four model-runs share dataset, contracts, and rubric, so cross-run agreement does not deliver the independence a confirmatory test would require. The exploratory signals are inputs to follow-on design (§6.2), not substitutes for it.

5.3 The score-rubric ceiling is a structural finding, not a side issue

The location sub-score’s empirical near-saturation (§3.7) is the binding constraint on confirmatory power in this study. The rubric assigns 0, 1, or 2 points to the location component of a T2 verdict; the saturation rate at the 2-point ceiling, documented in the empirical distribution of §3.7, leaves almost no headroom for C4 to outscore C1+ on this component within a paired comparison. The compressed effective Wilcoxon n on H1/T2—14 against $n_{\text{pairs}} = 30$ (§4.1)—is the direct consequence: 16 of 30 paired differences tie at zero, and the moderate point estimate must clear Holm correction with that compressed denominator.

This is a property of the as-implemented T2 rubric, not of the dataset or the models, and the distinction determines the path forward. Scaling n from 30 to 50 or 100 would not change the saturation rate; it would multiply tied paired differences in the same proportion, leave effective n in the same fraction, and leave the Holm-corrected α -budget no easier to meet. n expansion is necessary, but it is not sufficient and not first. The follow-on (§6.2) must refine the location sub-score—finer ordinal categories, or a continuous-valued T2 metric supplementing the composite—before expanding n . We state this as a methodological finding with reuse value beyond the present study: in any composite-rubric, paired-comparison design where one sub-score saturates against its ceiling, n expansion alone will be a costly and ineffective response to underpowering.

5.4 Implications for the long-term Lumen vision

Constitution v0.2.2 §3 specifies a reassessment threshold for the canonical-core-over-surface-text bet—the design wager that a structured canonical core IR with text and LLM-facing notations as projections will outperform surface-text representations for code reasoning. The threshold: an outcome of $C1+ \geq C4$ on all primary metrics across both confirmatory models would trigger

formal reassessment. The observed outcome on $H1/T2 \rightarrow C4 > C1+$ in direction, moderate effect size, failing Holm correction at $m = 9$ —does not meet that threshold and therefore does not trigger reassessment.

It also does not constitute confirmatory support for the bet. The point estimate is consistent with the bet’s direction; the statistical confirmation it would require has not been delivered. The empirical motivation for the bet is intact at the level of point estimates and weakened at the level of statistical confirmation. T1 and T3 have since been collected (§4.3–§4.4) and did *not* supply the missing confirmation: H1 is a clean null on T1 and ceiling-censored on T3. The project continues at the architectural level pending rubric refinement (§5.3) and a properly powered, ceiling-free confirmatory re-test (§6.2). Neither overclaim nor write-off is warranted; both would misrepresent the data.

5.5 Threats to validity (consolidated)

Several threats remain on the table and bound what the present results support.

- **Power.** $n_{\text{pairs}} = 30$ with effective Wilcoxon $n = 13$ –14 after rubric-induced ties (§3.6, §3.7) bounds power on every confirmatory cell.
- **Phase-2 power analysis not produced.** Constitution §16 (Open Question 4) called for a Phase-2 pilot power analysis or effect-size sanity check before fixing n_{pairs} ; this artifact was not produced (§3.2, Appendix E), and the rubric-induced compression of effective n was therefore identified post hoc rather than pre-confirmatory.
- **Family completeness and measurement limits.** The nine-cell family is now complete (T1/T3 added, §4.3–§4.4), but two of the three tasks are measurement-limited: T2 by a low-end rubric ceiling and T3 by a high-end solve-rate ceiling (93.7% of T3 scores at 1.0, effective $n \approx 5$). The one ceiling-free task, T1, returns a clean null for H1. The T1 collection hit a transient 27/300-cell gap (`claude-opus-4-6`; Anthropic credit exhaustion mid-run) that was recovered on resume after a credit top-up; the final T1 dataset is complete (300/300).
- **Dataset domain skew.** Approximately two-thirds (20 of 30) of the retained functions are list and array operations; results may not generalize uniformly to arithmetic, parsing, or string-heavy domains.
- **Author-reviewed contracts.** Single-evaluator residual variance enters the rubric inputs. For H1 ($C4$ vs $C1+$) the same reviewed contracts are present in both conditions, so reviewer-side variance cancels in the paired comparison; for H3 ($C1+$ vs $C1$) the contracts are present on $C1+$ only, and any systematic reviewer bias does not cancel across the pair.
- **Two providers, two generations.** The frontier-LLM landscape is not fully covered.
- **Non-recursive single-function scope.** A Constitution-mandated 90-day scope restriction.
- **Information parity is operational, not absolute.** The residual asymmetry note in §3.3 stands; round-trip $C4 \rightarrow C1+ \rightarrow C4'$ validation is journal-version follow-on work.

6 Conclusion

6.1 Summary

We proposed a methodology for information-parity testing of code-representation effects on frozen frontier LLMs and instantiated it on T2 bug detection. On the pre-registered primary endpoint

the result is *direction-consistent but Holm-non-significant*: the point estimate $r_{\text{rb}} = +0.581$ lies in the pre-registered direction, but the corrected $p = 0.205$ at $m = 9$ does not reject H_0 . Exploratory analyses find the H1 sign positive in all four model-runs (including the successor pair under a relaxed sampling regime; §3.6) and a systematic H3 sign inversion across the same four model-runs—reported under exploratory labeling, not promoted to confirmatory claims. Post-hoc inspection identifies a scoring-rubric ceiling as the structural source of confirmatory underpowering.

6.2 Immediate follow-on

Several concrete steps follow from the present results. They are listed in priority order, with the first item as the binding constraint to address before any dataset expansion.

- **Refine the T2 scoring rubric** to address the location-sub-score saturation (§3.7, §5.3). This is the binding constraint on confirmatory power and precedes n expansion.
- **Collect T1 and T3 data** under a separate confirmatory pre-registration, so that the six insufficient-data cells of the nine-test family (§4.2) become testable.
- **Implement the $C4 \rightarrow C1+ \rightarrow C4'$ round-trip validation** (§3.3, Appendix D) to quantify the information-parity residuals beyond the operational definition adopted here.
- **Back-port the bootstrap CI** into the frozen `analyze_confirmatory.py` via a structurally equivalent re-implementation, now that the sibling CI implementation has been verified against the frozen point estimates (§3.6, Appendix E).
- **Expand n** (e.g. to 50) under separate pre-registration—*after* rubric refinement, not before (§5.3).
- **Broaden dataset domain coverage** to dilute the list-operations skew documented in §5.5.

6.3 Long-term

The study tests a precondition for the broader Lumen vision — a canonical core IR with text and LLM-facing notations as projections: namely, that representation form materially affects external-agent reasoning. The direction-consistent confirmatory point estimate, together with the exploratory agreement of the H1 sign across all four model-runs (the successor pair under the relaxed sampling regime of §3.6, not a cross-generation replication), keeps that precondition *plausible*; it does not establish it. A definitive empirical case awaits T1 and T3 collection and rubric refinement; the canonical-core-over-surface-text bet (§5.4) remains active, and is reassessed only on stronger contrary evidence than the present data provide.

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Appendices

A Retained 30 func_ids and domain categorization

The 30 retained non-recursive Python functions used for the confirmatory and successor T2 runs are listed below in the order recorded by `input_summary.func_ids_used`.

Table 8. Retained 30 func_ids with their domain categorization in {arithmetic, list_operations, string_processing, statistics, control_flow}. Categories derived from each function’s signature and docstring in `data/functions/raw/`.

#	func_id	Domain
1	antidiagonals	list_operations
2	batch_list	list_operations
3	camel_to_snake	string_processing
4	chunk_on_change	list_operations
5	count_true_segments	list_operations
6	dense_rank	statistics
7	equal_width_buckets	statistics
8	find_balanced_spans	control_flow
9	first_index_of_max	list_operations
10	frequency_table	statistics
11	group_and_aggregate	statistics
12	group_by_key	list_operations
13	longest_plateau	list_operations
14	max_subarray_bounds	arithmetic
15	merge_intervals	list_operations
16	merge_sorted	list_operations
17	peak_valley_indices	list_operations
18	remove_adjacent_dups	list_operations
19	rle_encode	list_operations
20	rotate_list	list_operations
21	segments_above_threshold	list_operations
22	sliding_window_max	list_operations
23	sorted_list_intersection	list_operations
24	spiral_order	list_operations
25	stable_partition	list_operations
26	strided_windows	list_operations
27	tokenize_arithmetic	string_processing
28	top_k_by	statistics
29	two_sum_sorted_pairs	list_operations
30	welford_running_stats	statistics

Aggregate. arithmetic 1, list_operations 20, string_processing 2, statistics 6, control_flow 1; total 30.

Curation rule. A static recursion-exclusion check at the dataset curation stage rejects any function with self-recursion or mutually recursive helpers; the 90-day study scope explicitly restricts to non-recursive code (Constitution §17.5).

B Pre-registration timeline and key commit hashes

Pre-registration : Lumen Design Constitution v0.2.2

```

commit 6b83bca
(docs/design-constitution.tex was
committed at repository scaffolding
and never edited thereafter; this
is the sole commit touching that
file in the git log)

Analysis machinery freeze      : 9217a26
                               phase7b: freeze canonical execution
                               state for full T2 confirmatory v2 rerun

Confirmatory closure          : 10beff2
                               phase7c: close confirmatory T2
                               full_t2_confirmatory_v2 authoritative

Successor plan registration    : 9c91347
                               phase7d: register T2 frontier
                               successor replication plan

Exploratory tooling           : 865fbe0
                               phase7e: close T2 frontier successor
                               exploratory replication

Phase 7e priority 2           : 54e722c
                               phase7e-p2: implement
                               analyze_exploratory.py + tests

```

C Full E1–E3 results, confirmatory v2 and successor runs

Table 9. Full E1–E3 cumulative-decomposition results, confirmatory v2 run (`full_t2_confirmatory_v2`, 2-model averaged) and successor run (`t2_frontier_successor_replication`, 2-model averaged). *Exploratory analysis.* Each row pairs the cumulative step (C2 vs C1, C3 vs C2, C4 vs C3) on T2.

Run	Step	n_{pairs}	W	raw p	r_{rb}	95% CI	$n_{+}/n_{-}/n_{=}$
v2	E1 (C2 vs C1)	30	75.0	0.173	+0.250	$[-0.273, +0.732]$	9/6/15
v2	E2 (C3 vs C2)	30	22.0	0.828	−0.200	$[-0.778, +0.429]$	4/6/20
v2	E3 (C4 vs C3)	30	22.5	0.363	+0.250	$[-0.500, +1.000]$	5/3/22
successor	E1 (C2 vs C1)	30	30.0	0.915	−0.341	$[-0.806, +0.238]$	5/8/17
successor	E2 (C3 vs C2)	30	17.5	0.383	+0.250	$[-0.667, +1.000]$	4/3/23
successor	E3 (C4 vs C3)	30	5.0	0.844	−0.333	$[-1.000, +1.000]$	2/3/25

Notes. Raw p -values are one-sided “greater” matching the pre-registered direction of each cumulative step. W is the Wilcoxon signed-rank statistic. Confidence intervals are 95% percentile bootstrap (seed 20260427, 10 000 resamples). The high tie counts ($n_{=} \in \{15, 17, 20, 22, 23, 25\}$) are the same effective- n compression phenomenon documented for the confirmatory family in §3.7.

D Information-parity round-trip validation

Method

To quantify how much of the Core IR (C4) survives the projection to the information-parity condition (C1+), we run a round trip $C4 \rightarrow C1+ \rightarrow C4'$ on all 30 retained confirmatory functions and compare C4 against the re-derived C4'. C1+ is produced by the same projection used in the experiment (`src/pipeline/ir_to_annotated_text.py`). C4' is re-derived from the C1+ *text only*: the AST by re-parsing the C1+ source, the contract clauses by parsing the structured docstring block (a docstring→contract parser written for this purpose, since the production lowering reads the authoritative reviewed contract rather than the projected text), and the types from the C1+ annotations. Comparison normalization is specified in `docs/roundtrip-comparison-spec.md`; the implementation is `src/experiment/roundtrip.py` and the per-function and aggregate outputs are under `results/analysis/exploratory/roundtrip/`. This is an exploratory measurement, not a confirmatory test.

Results

Table 10 reports per-metric recovery over the 30 functions.

Table 10. $C4 \rightarrow C1+ \rightarrow C4'$ recovery over the 30 retained confirmatory functions (*Exploratory analysis*). Retention is the fraction of the C4 quantity recovered in C4'; see `docs/roundtrip-comparison-spec.md` for the normalization of each metric.

Metric	Mean	Min	Max
Node-kind retention	0.998	0.962	1.000
Type-annotation retention	0.964	0.600	1.000
Parent-child (edge) retention	0.987	0.885	1.000
Cross-reference retention	1.000	1.000	1.000
Contract preconditions	1.000	1.000	1.000
Contract postconditions	1.000	1.000	1.000
Contract invariants	1.000	1.000	1.000
Canonical-hash agreement	0/30 functions (exact AST match)		

The propositional content the experiment relies on is recovered essentially in full: all three contract-clause families and the cross-reference structure among clauses are recovered at 100%, node kinds at 99.8%, type annotations at 96.4%, and parent-child structure at 98.7%. What is *not* recovered is byte-exact AST identity: 0/30 functions round-trip to an identical AST, because C1+ introduces a structured docstring node and does not preserve the IR's syntactic child-ordering or node-level metadata.

Interpretation

These rates bear on the information-parity claim but do not settle the study's question, and must not be read in a confirmatory tone. High recovery means C1+ is a high-fidelity, information-preserving projection of C4 rather than a lossy summary: the contract content and cross-references that the reasoning task depends on survive the projection intact. *If* C4 is nonetheless found to outperform C1+ in a future adequately-powered comparison, this high recovery *strengthens*—but does not establish—the reading that representational structure itself, not differing information content, drives the effect; the residual non-recovered structure (byte-exact AST identity, 3.6% of type annotations, 1.3% of edges) remains a candidate confound. Conversely, were recovery low on the content metrics, the parity claim would be weakened and the C4-vs-C1+ contrast would be partly confounded by information loss; that is not what we observe here for the content metrics,

but the never-exact AST identity is stated plainly as the boundary of the parity claim. No causal or confirmatory conclusion is drawn from this exploratory measurement.

E Reproducibility manifest

Repository

```
git@github.com:WhiteStoneTak/Lumen.git  
HEAD at submission preparation: fca5e85
```

Key commit hashes (also Appendix B)

```
Pre-registration          : Constitution v0.2.2  
                          (committed before data collection)  
  
Analysis machinery freeze : 9217a26  
Confirmatory closure     : 10beff2  
Successor plan registration : 9c91347  
Exploratory tooling      : 865fbe0  
analyze_exploratory.py   : 54e722c
```

Run wall-clocks

```
full_t2_confirmatory_v2      : 36 min 13 s  
t2_frontier_successor_replication : 45 min 25 s
```

API endpoint snapshot dates

```
Data collection day (both runs) : 2026-04-27 (local time)  
  Confirmatory : 2026-04-27 00:44 -- 01:21 (36 min 13 s)  
  Successor    : 2026-04-27 09:09 -- 09:55 (45 min 25 s)  
Requested model_ids (logical, as passed to the provider SDKs)  
  Confirmatory : claude-opus-4-6, gpt-5.4  
  Successor    : claude-opus-4-7, gpt-5.5  
Provider-side dated snapshot identifiers (e.g., Anthropic dated  
suffix, OpenAI system_fingerprint) were not captured at  
collection time and are not recoverable from the saved response  
archives, which store only model_id, response_text, prompt_tokens,  
total_tokens, and error. We declare this as a provenance  
limitation; future runs should record these fields at the SDK  
call site.
```

Raw response archives

```
Confirmatory : results/runs/full_t2_confirmatory_v2/responses/  
Successor    : results/runs/t2_frontier_successor_replication/responses/  
(Score JSONs alongside, under ../scores/.)
```

Pre-confirmatory dataset-sufficiency audit (referenced by §3.1, §5.5)

```
Path          : results/analysis/t2_dataset_sufficiency_audit_2026-04-02.json  
Schema        : t2-dataset-sufficiency-audit-v1  
Audit date    : 2026-04-02 (three weeks before full_t2_confirmatory_v2  
                      collection on 2026-04-27)  
Key finding   : dataset_sufficiency_finding records
```



```

    "minimum_line_reached": true,
    "sufficiency_proven": false, and
    "evidence_for_30_vs_50":
        "No in-repo pilot power analysis artifact or
        effect-size sanity-check artifact determining
        30 vs 50 sufficiency was found."
Decision      : branch_decision records that confirmatory execution
                proceeded at the constitutional minimum line of
                n = 30 (wave2 pool exhausted), with rubric-induced
                effective-n compression identified post hoc from
                confirmatory data (g3.7, g5.3).

```

Bootstrap CI scratch-script provenance (confirmatory effect sizes)

```

Script        : scratch/bootstrap_ci_confirmatory.py
Results JSON  : scratch/bootstrap_ci_results.json
RNG seed      : 20260427
Resamples     : 10,000
Method        : percentile bootstrap on rank-biserial r
Verification  : point estimates re-computed by the script and
                matched analyze_confirmatory.py output to <0.001
                on r and to identical W and raw p-values on the
                Wilcoxon decisions.

```

The frozen analyze_confirmatory.py does not compute CI; the scratch script is a sibling implementation that adds CI reporting only and does not modify any pre-registered hypothesis-testing decision.

Figure provenance

```

F1 (rendered as Figure 2; H1/T2 per-model forest)
Script      : paper/figures/scripts/make_f1_f2_forest.py
Source      : results/analysis/exploratory/
              full_t2_confirmatory_v2/exploratory_results.json
              results/analysis/exploratory/
              t2_frontier_successor_replication/exploratory_results.json
              (per_model_split_h1_h2_h3; H1, T2 entries)
Output      : paper/figures/F1.pdf
Recompute:  python3 paper/figures/scripts/make_f1_f2_forest.py
No statistics are recomputed; r and 95% bootstrap CI are read
verbatim from the analysis artifacts and plotted as-is.
Script asserts each row's r and CI against the values in
Tables 4-5 of the paper.

```

```

F2 (rendered as Figure 3; H3/T2 per-model forest)
Same script and same source files as F1
(per_model_split_h1_h2_h3; H3, T2 entries).
Output      : paper/figures/F2.pdf
Script asserts each row against Table 6, including the collapsed
CI [-1.000, -1.000] on the claude-opus-4-7 row.

```

F3 (rendered as Figure 4; T2 score distribution histograms)
Script : paper/figures/scripts/make_f3_histograms.py
Source : results/runs/full_t2_confirmatory_v2/scores/*.json
 (scorer-result-v1; 300 records)
Output : paper/figures/F3.pdf
Descriptive count over the frozen confirmatory run (no
hypothesis test). Left panel = per-model raw composite over
300 records; right panel = per-(function, condition) 2-model
averaged composite over the $30 \times 5 = 150$ cells.
Script asserts the left-panel counts against section 3.7
({0:19, 1:89, 2:192, 3:0}; total 300) and that the right-panel
2.5 and 3.0 supports are empty.

F4 (rendered as Figure 1; 5-condition schematic)
No data source; drawn inline in paper/main.tex as TikZ.